Introduction to Algebra

AQA KS3 · Year 7 · Foundation

A letter stands for a number we do not know yet. Simplify each expression by collecting like terms. Show your working in your workbook.

Method

1 Find the like terms (same letter)   2 Group them, keeping each sign   3 Add or subtract the number parts

4 Letters first, then the number   5 Check by substituting a value

Common mistakes

- 3x means 3 × x. It is not the digits 3 and x written side by side.
- You can only collect like terms: 3x + 2x = 5x, but 3x + 2 stays as 3x + 2.
- x on its own means 1x, so x + x = 2x — not x^2.
- Multiplying is different from adding: $x \times x = x^2$, but x + x = 2x.

Worked example

Simplify: 2x + 3 + x + 1

Step 1 — Like terms: 2x and x are like; 3 and 1 are like.

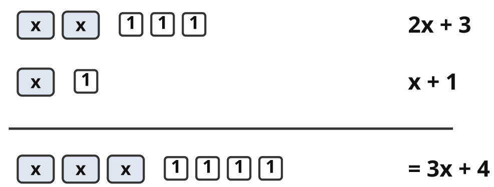
Step 2 — Group them: (2x + x) + (3 + 1)

Step 3 — Combine the x terms: 2x + x = 3x

Step 4 — Combine the numbers: 3 + 1 = 4

Answer: 3x + 4 (check x = 5: 2×5+3+5+1 = 19 and 3×5+4 = 19)

Visual aid — collecting like terms



Tiles of the same kind join together: 2x + 3 and x + 1 give 3x + 4.

Questions

Simplify each expression — show your working in your workbook

Q1 of 8 [1]

Simplify: $a + a + a + a$

Q2 of 8 [1]

Simplify: $5x + 2x$

Q3 of 8 [2]

Simplify: $5x + 2x - 3x$

Q4 of 8 [2]

Simplify: $7y + 4 + 2y + 5$

Q5 of 8 [2]

Simplify: $8m + 3n - 2m + 5n$

Q6 of 8 [2]

A pencil costs p pence. Write an expression for the cost of 4 pencils and a 30p rubber.

Q7 of 8 [3]

Expand and simplify: $3(x + 2) + 2x$

Q8 of 8 [3]

A rectangle has length $(x + 3)$ and width x. Write a simplified expression for its perimeter.